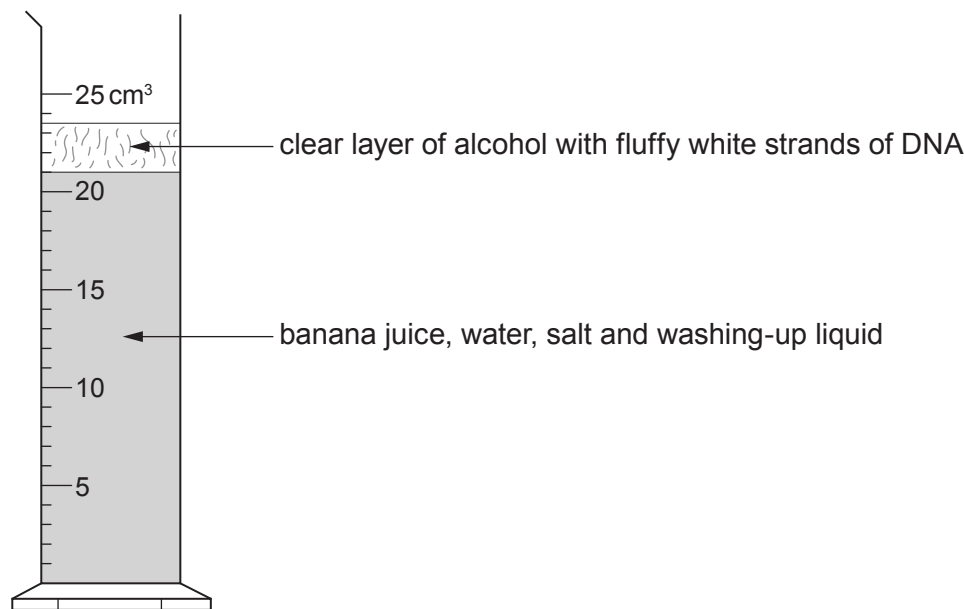


4. A group of year 11 students extracted DNA from bananas in the school laboratory. The result of this extraction process is shown in **Image 4.1**.

Image 4.1



The students collected the DNA from the measuring cylinder and took it to the local university where members of the biology staff had agreed to supervise the students whilst they analysed its base content. The readings they obtained are shown in **Image 4.2**.

Image 4.2

DNA base analysis. Source – banana	
Adenine	33%
Cytosine	17%
Guanine	17%
Thymine	33%



Examiner
only

(a) State the ratio of:

[1]

(i) l. adenine to thymine

.....

II. guanine to cytosine.

(ii) Explain these ratios by referring to the arrangement of these bases in a molecule of DNA. [2]

[2]

(iii) If the students had repeated the experiment with strawberries would the base analysis be the same as that of bananas? Explain your answer.

[2]

(b) Explain the role of the bases in protein synthesis.

[2]

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7

